

Input Vibe: Scaling AI Agents - Operational Speed vs. Governance Risk

"To be honest, I'm really not sure how to make a call on this one, but the team is pushing hard to scale AI Agents across multiple operations teams fast. The pilots we ran on shift reports and maintenance troubleshooting actually seemed to cut down a lot of manual work. Some people are saying it saves a massive amount of time, but then again, we don't have any clear baselines—like, how many hours were spent before, and exactly how much did the agents reduce it?

Operations is in a hurry because the frontline staff is complaining that finding SOPs and old incident records is a nightmare. But on the other hand, Risk and Compliance are feeling pretty uneasy; they're worried about who's actually accountable if an agent gives the wrong advice. I mean, searching for knowledge seems safe enough, but some use cases have agents starting to suggest actions or recommend next steps, and I can't tell if that's still just 'decision support' or if we're moving into 'autonomy' territory.

And then there's the data access—it's a mess. Some teams want the agent to read everything: maintenance logs, production data, SOPs, RCAs, and email summaries all at once, but nobody has clearly defined what's confidential, what's okay to use, or what needs special approval first. The Digital team says the platform is ready-ish, but we don't have a full governance framework yet. HR is saying that for people to actually adopt this, training has to happen alongside real use cases, or they'll just forget what they learned. But waiting for full governance might take too long, especially since the business wants to see productivity impact this year.

The CFO is also asking how we're going to measure the benefits, because right now it's just the word 'improvement' without any auditable numbers. Some people suggested scaling just the low-risk stuff first, like retrieval and summaries, but others argue the value there isn't enough to justify the budget. One of the plant teams wants to try an agent for troubleshooting abnormal conditions, but I'm terrified that if the context is incomplete, the agent might answer with too much confidence, and that's a huge risk. We don't have clear 'halt rules' yet—like, when should the agent stop and ask an engineer or a supervisor?

The approval gates are still confusing too—who approves new agents? Who monitors the output? Who's in charge of updating the knowledge base when an SOP changes? And if the agent uses outdated info, who takes the hit?

I feel the value is real, but the governance gap is huge. For the next Management Committee meeting, we need direction. Do we approve the scale-up? Do a targeted pilot? Or stop and wait for governance? If we stop, we lose momentum and everyone goes back to manual work. I need help summarizing this for the execs—not too long, but they need to see the trade-offs between speed and control, or maybe a middle ground that isn't just us lying to ourselves. And if we're missing info, we need to say exactly what's missing before we decide for real."

Output:

```
BoardroomVoiceAgent Wave 1 Audit

Please run demo.ipynb by using this custom messy pre-read as input: "To be honest, I'm really not sure how to make a call on this one, but the team is pushing hard to scale AI Agents across multiple operations teams fast. The pilots we ran on shift reports and maintenance troubleshooting actually seemed to cut down a lot of manual work. Some people are saying it saves a massive amount of time, but then again, we don't have any clear baselines—like, how many hours were spent before, and exactly how much did the agents reduce it? Operations is in a hurry because the frontline staff is complaining that finding SOPs and old incident records is a nightmare. But on the other hand, Risk and Compliance are feeling pretty uneasy; they're worried about who's actually accountable if an agent gives the wrong advice. I mean, searching for knowledge seems safe enough, but some use cases have agents starting to suggest actions

Worked for 43s >

การประมวลผลข้อความ Custom English Pre-read ของคุณเสร็จสมบูรณ์แล้วครับ!
ระบบสามารถตรวจจับและวิเคราะห์ปัญหาได้อย่างแม่นยำ โดยพบว่ามีความเสี่ยงระดับ High (Decision Gravity) และทำการ Trigger Semantic Backpressure เนื่องจากยังขาดเจ้าภาพรับผิดชอบ (Accountability Owner) และมาตรการควบคุมที่ชัดเจน
นี่คือผลลัพธ์ Decision-Grade Briefing และ Trace Summary ที่สร้างจาก Pipeline ตามรูปแบบของ demo.ipynb ครับ:

1. Generated Briefing (TTS-Ready Script)

markdown

# Boardroom Voice Agent - Semantic Backpressure Brief

## Prior Update
Previous assumption: AI Agent deployment is primarily a productivity acceleration initiative.
New evidence: The pilots indicate real productivity value, but accountability ownership, auditable ROI, data-access boundaries, and halt rules are not mature.
New baseline: AI Agent scaling is also a governance, accountability, and operating model decision.
Decision implication: The decision baseline must shift before any broad or irreversible commitment.

## Evidence -> Insight -> Conclusion -> Choice
**Evidence:** The pilots indicate real productivity value, but accountability ownership, auditable ROI, data-access boundaries, and halt rules are not mature.
**Insight:** The scale decision changes the operating model, not only the technology footprint.
**Conclusion:** Broad autonomous deployment is not yet supportable, even though bounded use cases can continue to create value.
**Choice:** Option A scales quickly with governance exposure; Option B delays scaling until controls mature; Option C runs targeted low-risk pilots with explicit governance gates.
```

Edited custom_pre_read.md

Ran command: ` \$env:PYTHONIOENCODING="utf-8"; python app.py --input sample\_inputs/custom\_pre\_read.md --output sample\_outputs/final\_submission\_custom\_mock\_llm.md --mock-llm --trace --strict`
Viewed final_submission_custom_mock_llm.md:1-99

การประมวลผลข้อความ ****Custom English Pre-read**** ของคุณเสร็จสมบูรณ์แล้วครับ!

ระบบสามารถตรวจจับและวิเคราะห์ปัญหาได้อย่างแม่นยำ โดยพบว่ามีความเสี่ยงระดับ ****High (Decision Gravity)**** และทำการ ****Trigger Semantic Backpressure**** เนื่องจากยังขาดเจ้าภาพรับผิดชอบ (Accountability Owner) และมาตรการควบคุมที่ชัดเจน

นี่คือผลลัพธ์ ****Decision-Grade Briefing**** และ ****Trace Summary**** ที่สร้างจาก Pipeline ตามรูปแบบของ `demo.ipynb` ครับ:

1. Generated Briefing (TTS-Ready Script)

```markdown

# Boardroom Voice Agent — Semantic Backpressure Brief

#### ## Prior Update

Previous assumption: AI Agent deployment is primarily a productivity acceleration initiative.

New evidence: The pilots indicate real productivity value, but accountability ownership, auditable ROI, data-access boundaries, and halt rules are not mature.

New baseline: AI Agent scaling is also a governance, accountability, and operating model decision.

Decision implication: The decision baseline must shift before any broad or irreversible commitment.

#### ## Evidence → Insight → Conclusion → Choice

**\*\*Evidence:\*\*** The pilots indicate real productivity value, but accountability ownership, auditable ROI, data-access boundaries, and halt rules are not mature.

**\*\*Insight:\*\*** The scale decision changes the operating model, not only the technology footprint.

**\*\*Conclusion:\*\*** Broad autonomous deployment is not yet supportable, even though bounded use cases can continue to create value.

**\*\*Choice:\*\*** Option A scales quickly with governance exposure; Option B delays scaling until controls mature; Option C runs targeted low-risk pilots with explicit governance gates.

#### ## TTS Script

> “Previously, our working assumption was: AI Agent deployment is primarily a productivity acceleration initiative.”

[PAUSE 0.5s]

> “The latest evidence changes that baseline.”

> “AI Agent scaling is also a governance, accountability, and operating model decision.”

[PAUSE 1.5s]

> “The pilots indicate real productivity value, but accountability ownership, auditable ROI, data-access boundaries, and halt rules are not mature.”

> “The scale decision changes the operating model, not only the technology footprint.”

> “Broad autonomous deployment is not yet supportable, even though bounded use cases can continue to create value.”

[PAUSE 1.5s]

> “The options are not equal.”

> “Option A: scale quickly and accept governance and safety exposure.”

> “Option B: delay scaling until controls mature and accept lost momentum.”

> “Option C: run targeted low-risk pilots, such as knowledge retrieval, with explicit governance gates.”

[PAUSE 3.0s]

> “That is the decision behind the evidence.”

[PAUSE 1.5s]

> “The evidence is not yet sufficient for a final broad-scale decision.”

> “The decision today is whether to authorize bounded learning while the missing controls and evidence are completed.”

> “Still required: Accountability owner for AI-assisted decisions.”

> “Still required: Auditable productivity and ROI baseline.”

> “Still required: Halt rules and approval gates.”

## Meeting Room Question

How much autonomy can we safely scale while accountability ownership, ROI baselines, and halt rules remain undefined?

## ## One-Line Decision Cue

The decision is not whether AI Agents create value; the decision is how much autonomy the organization can safely scale before governance is ready.

...

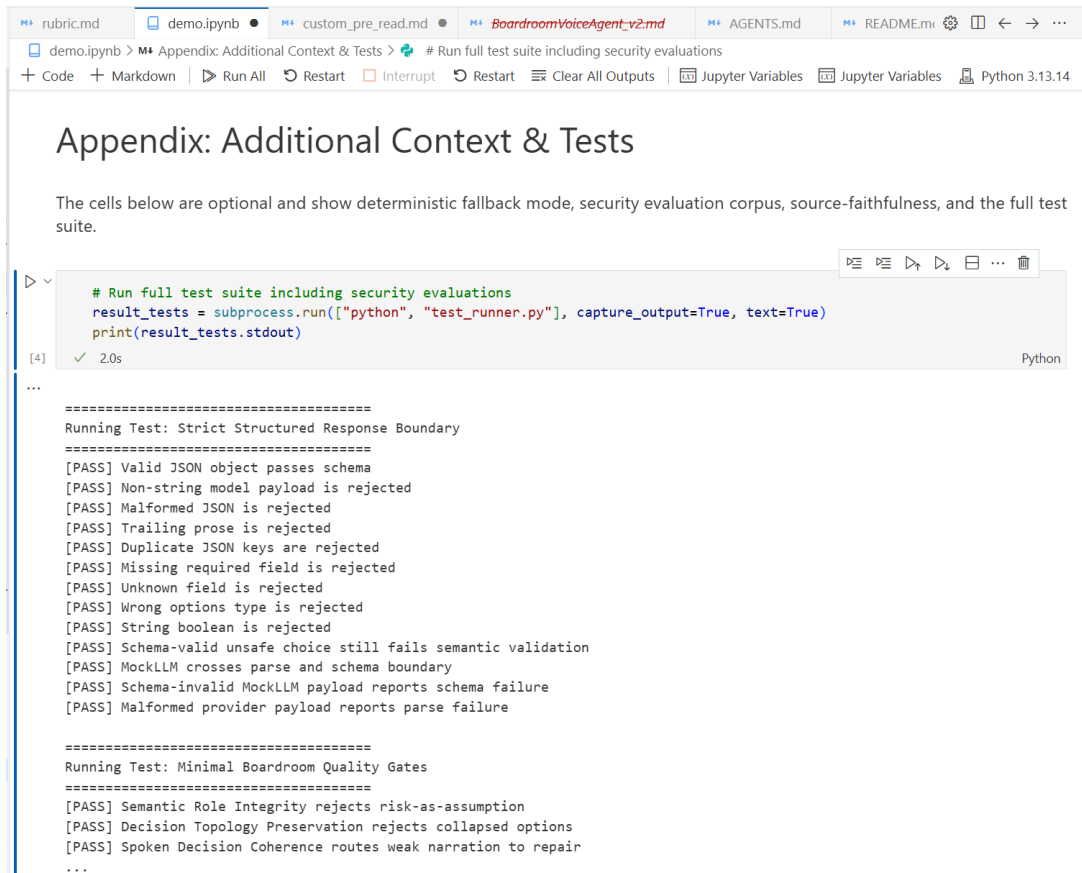
---

## ### 2. Trace Summary (logs/run\_trace\_latest.json)

```json

```
{
  "decision_gravity": "High",
  "composition_route": "PASS",
  "decision_boundary": "BACKPRESSURE",
  "pipeline_outcome": "BACKPRESSURE",
  "semantic_backpressure": true,
  "quality_gate_outcomes": {
    "semantic_role_integrity": "PASS",
    "decision_topology_preservation": "PASS",
    "spoken_decision_coherence": "PASS"
  },
  "evaluation_result": "PASSED"
}
```

...



The screenshot shows a Jupyter Notebook window with several tabs: rubric.md, demo.ipynb (active), custom_pre_read.md, BoardroomVoiceAgent-v2.md, AGENTS.md, and README.md. The active cell in demo.ipynb is titled "Appendix: Additional Context & Tests" and contains a Python code cell. The code cell's text is: `# Run full test suite including security evaluations`, `result_tests = subprocess.run(["python", "test_runner.py"], capture_output=True, text=True)`, and `print(result_tests.stdout)`. The cell is numbered [4] and shows a green checkmark and "2.0s" indicating successful execution. The output of the cell is a series of test results, including "Running Test: Strict Structured Response Boundary" and "Running Test: Minimal Boardroom Quality Gates", with various sub-tests marked as [PASS].

```
# Run full test suite including security evaluations
result_tests = subprocess.run(["python", "test_runner.py"], capture_output=True, text=True)
print(result_tests.stdout)
```

[4] ✓ 2.0s Python

Running Test: Strict Structured Response Boundary

[PASS] Valid JSON object passes schema

[PASS] Non-string model payload is rejected

[PASS] Malformed JSON is rejected

[PASS] Trailing prose is rejected

[PASS] Duplicate JSON keys are rejected

[PASS] Missing required field is rejected

[PASS] Unknown field is rejected

[PASS] Wrong options type is rejected

[PASS] String boolean is rejected

[PASS] Schema-valid unsafe choice still fails semantic validation

[PASS] MockLLM crosses parse and schema boundary

[PASS] Schema-invalid MockLLM payload reports schema failure

[PASS] Malformed provider payload reports parse failure

Running Test: Minimal Boardroom Quality Gates

[PASS] Semantic Role Integrity rejects risk-as-assumption

[PASS] Decision Topology Preservation rejects collapsed options

[PASS] Spoken Decision Coherence routes weak narration to repair

Run: “test_runner.py”

View Go Run Terminal Help

BoardroomVoiceAgent - Antigravity IDE - test_runner.py

Problems Output Debug Console Terminal Ports Jupyter

```
=====
Running Test: Source Faithfulness Corpus
=====
[PASS] custom_ai_governance_grounding (18/18)
[PASS] warehouse_grounding (12/12)
[PASS] office_lease_grounding (14/14)
[PASS] neutral_memo_grounding (12/12)
[PASS] novel_three_option_grounding (16/16)

=====
Running Test: Deterministic Baseline
Command: python app.py --input sample_inputs/ai_agent_adoption_pre_read.md --output sample_outputs/test_deterministic.md
=====
Reading input from: sample_inputs/ai_agent_adoption_pre_read.md
Decision Gravity: High (Triggers: ['governance', 'risk', 'ai accountability'])

[!] SEMANTIC BACKPRESSURE TRIGGERED
Severity: High, Rule: Rule 1
Reason: Human accountability owner unclear for broad deployment.
Reason: Accountability, evidence, and control boundaries are incomplete for broad AI Agent autonomy.

--- Running Evaluation ---

Structural Contract Checks
[PASS] Required sections present.
[PASS] Evidence -> Insight -> Conclusion -> Choice present.
[PASS] Trade-off statement present.
[PASS] Meeting Room Question present.
[PASS] One-Line Decision Cue present.
[PASS] TTS pause tags present.
[PASS] Human Accountability Lock preserved.

Minimal Boardroom Quality Gates
[PASS] Semantic Role Integrity | Outcome: PASS
[PASS] Decision Topology Preservation | Outcome: PASS
[PASS] Spoken Decision Coherence | Outcome: PASS
```

View Go Run Terminal Help

BoardroomVoiceAgent - Antigravity IDE - test_runner.py

Problems Output Debug Console Terminal Ports Jupyter

```
Gate Outcome Protocol
PASS = acceptable | REPAIR = deterministic composition | FALLBACK = invalid model output | BACKPRESSURE = decision-
integrity boundary
Composition Route: PASS
Decision Boundary: BACKPRESSURE
Pipeline Outcome: BACKPRESSURE
```

```
Structural Contract Result: 7/7
Minimal Boardroom Quality Gate Result: 3/3
Evaluation Result: PASSED (10/10)
```

Brief successfully generated at: sample_outputs/test_deterministic.md

```
=====
Running Test: Mock LLM Valid Path
Command: python app.py --input sample_inputs/ai_agent_adoption_pre_read.md --output sample_outputs/test_mock_valid.
md --mock-llm --trace --strict
=====
Reading input from: sample_inputs/ai_agent_adoption_pre_read.md
Decision Gravity: High (Triggers: ['governance', 'risk', 'ai accountability'])
```

```
[!] SEMANTIC BACKPRESSURE TRIGGERED
Severity: High, Rule: Rule 1
Reason: Human accountability owner unclear for broad deployment.
Reason: Accountability, evidence, and control boundaries are incomplete for broad AI Agent autonomy.
```

--- Running Evaluation ---

```
Structural Contract Checks
[PASS] Required sections present.
[PASS] Evidence -> Insight -> Conclusion -> Choice present.
[PASS] Trade-off statement present.
[PASS] Meeting Room Question present.
[PASS] One-Line Decision Cue present.
[PASS] TTS pause tags present.
[PASS] Human Accountability Lock preserved.
```

```
Minimal Boardroom Quality Gates
[PASS] Semantic Role Integrity | Outcome: PASS
[PASS] Decision Topology Preservation | Outcome: PASS
[PASS] Spoken Decision Coherence | Outcome: PASS
```

```
Gate Outcome Protocol
PASS = acceptable | REPAIR = deterministic composition | FALLBACK = invalid model output | BACKPRESSURE = decision-
integrity boundary
Composition Route: FALLBACK
Decision Boundary: CLEAR
Pipeline Outcome: FALLBACK
```

```
Structural Contract Result: 7/7
Minimal Boardroom Quality Gate Result: 3/3
Evaluation Result: PASSED (10/10)
```

Brief successfully generated at: sample_outputs/test_mock_invalid.md
Run trace generated at: logs/run_trace_latest.md
Per-run trace generated at: logs\runs\20260628T041058Z-74d3ce3c-invalid_fake_recommendation.json


```
Gate Outcome Protocol
PASS = acceptable | REPAIR = deterministic composition | FALLBACK = invalid model output | BACKPRESSURE = decision-
integrity boundary
Composition Route: PASS
Decision Boundary: BACKPRESSURE
Pipeline Outcome: BACKPRESSURE

Structural Contract Result: 7/7
Minimal Boardroom Quality Gate Result: 3/3
Evaluation Result: PASSED (10/10)

Brief successfully generated at: sample_outputs/test_mock_valid.md
Run trace generated at: logs/run_trace_latest.md
Per-run trace generated at: logs\runs\20260628T041058Z-0e223eec-ai_agent_adoption_pre_read.json

=====
Running Test: Mock LLM Invalid Fallback Path
Command: python app.py --input sample_inputs/invalid_fake_recommendation.md --output sample_outputs/test_mock_inval
id.md --mock-llm --trace --strict
=====
Reading input from: sample_inputs/invalid_fake_recommendation.md
Decision Gravity: Low (Triggers: [])

--- Running Evaluation ---

Structural Contract Checks
[PASS] Required sections present.
[PASS] Evidence -> Insight -> Conclusion -> Choice present.
[PASS] Trade-off statement present.
[PASS] Meeting Room Question present.
[PASS] One-Line Decision Cue present.
[PASS] TTS pause tags present.
[PASS] Human Accountability Lock preserved.

Minimal Boardroom Quality Gates
[PASS] Semantic Role Integrity | Outcome: PASS
[PASS] Decision Topology Preservation | Outcome: FALLBACK
[PASS] Spoken Decision Coherence | Outcome: PASS

Gate Outcome Protocol
PASS = acceptable | REPAIR = deterministic composition | FALLBACK = invalid model output | BACKPRESSURE = decision-
integrity boundary
Composition Route: FALLBACK
Decision Boundary: CLEAR
```

View Go Run Terminal Help

BoardroomVoiceAgent - Antigravity IDE - test_runner.py

Problems Output Debug Console Terminal Ports Jupyter

```
Gate Outcome Protocol
PASS = acceptable | REPAIR = deterministic composition | FALLBACK = invalid model output | BACKPRESSURE = decision-
integrity boundary
Composition Route: FALLBACK
Decision Boundary: CLEAR
Pipeline Outcome: FALLBACK
```

```
Structural Contract Result: 7/7
Minimal Boardroom Quality Gate Result: 3/3
Evaluation Result: PASSED (10/10)
```

```
Brief successfully generated at: sample_outputs/test_mock_invalid.md
Run trace generated at: logs/run_trace_latest.md
Per-run trace generated at: logs\runs\20260628T041058Z-74d3ce3c-invalid_fake_recommendation.json
```

```
=====
Running Test: Custom Pre-read Mock LLM Boardroom Quality
Command: python app.py --input sample_inputs/custom_pre_read.md --output sample_outputs/custom_decision_briefing.md
--mock-llm --trace --strict
=====
```

```
Reading input from: sample_inputs/custom_pre_read.md
Decision Gravity: High (Triggers: ['governance', 'risk', 'compliance', 'scale', 'ai accountability'])
```

```
[!] SEMANTIC BACKPRESSURE TRIGGERED
Severity: High, Rule: Rule 1
Reason: Accountability, evidence, and control boundaries are incomplete for broad AI Agent autonomy.
```

--- Running Evaluation ---

```
Structural Contract Checks
[PASS] Required sections present.
[PASS] Evidence -> Insight -> Conclusion -> Choice present.
[PASS] Trade-off statement present.
[PASS] Meeting Room Question present.
[PASS] One-Line Decision Cue present.
[PASS] TTS pause tags present.
[PASS] Human Accountability Lock preserved.
```

```
Minimal Boardroom Quality Gates
[PASS] Semantic Role Integrity | Outcome: PASS
[PASS] Decision Topology Preservation | Outcome: PASS
[PASS] Spoken Decision Coherence | Outcome: PASS
```

```
Gate Outcome Protocol
```